

What Changed and What Could Have Changed? State-Change Counterfactuals for Procedure-Aware Video Representation Learning

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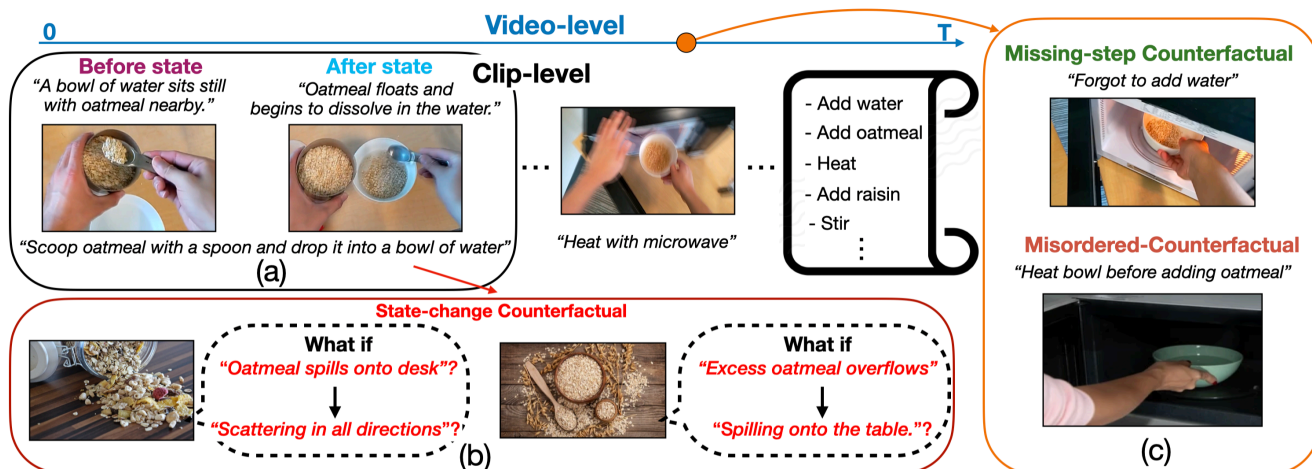


Figure 1. Illustration of “what change” (actual action-induced transformations) and “what could have changed” (hypothetical deviations) for both clip- and video-level procedures. (a) The clip narration “Scoop oatmeal with a spoon and drop it into a bowl of water” changes the scene’s **Before** and **After** states at the short-term clip-level. (b) Illustrates the hypothesized deviation, i.e., **State-change Counterfactual**, that could have happened and changed the scene differently at the clip level. (c) Illustrates the counterfactuals for video-level procedures, which depicts what could happen due to possible previous mistakes, for example, **Missing-step Counterfactual** and **Misordered Counterfactual**.

Abstract

Understanding a procedural activity requires modeling both how action steps transform the scene, and how evolving scene transformations can influence the sequence of action steps, even those that are accidental or erroneous. Existing work has studied procedure-aware video representations by modeling the temporal order of actions, but has not explicitly learned the state changes (scene transformations). In this work, we study procedure-aware video representation learning by incorporating state-change descriptions generated by Large Language Models (LLMs) as supervision signals for video encoders. Moreover, we gener-

ate state-change counterfactuals that simulate hypothesized failure outcomes, allowing models to learn by imagining unseen “What if” scenarios. This counterfactual reasoning facilitates the model’s ability to understand the cause and effect of each step in an activity. We conduct extensive experiments on procedure-aware tasks, including temporal action segmentation, error detection, action phase classification, frame retrieval, multi-instance retrieval, and action recognition. Our results demonstrate the effectiveness of the proposed state-change descriptions and their counterfactuals, and achieve significant improvements on multiple tasks. Code is available at <https://github.com/HCIS-Lab/counterfactual-video-pretrain>.

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1. Introduction

Procedural activities, such as following a cooking recipe [20, 24, 32, 37] or assembling a piece of furniture [49, 64], consist of sequences of interconnected steps that often happen in a specific and logical order to achieve a desired outcome. Understanding these activities from video data is essential for a variety of applications, including video retrieval [15, 22, 49, 84], intelligent collaborative agents [4, 27, 33, 79, 93], and robot learning from human demonstration [26, 30, 57, 65, 87, 89, 96, 105]. In contrast to general video action recognition that focuses on a single step in a short clip [7, 18, 21, 34, 77, 80], procedure-aware video understanding requires capturing both the “what changed” (actual action-induced state transformations) [41, 72], as shown in Figure 1 (a), and “what could have changed” (hypothetical deviations) [25, 58, 62, 101], as shown in Figures 1 (b) and (c). These two capabilities are important for understanding long-form procedures because the outcome of each action step can affect subsequent steps.

Procedure-aware video representation learning can enable various downstream long-form video tasks, such as temporal action segmentation [20, 32, 37, 64] and error detection [37, 64]. There have been many approaches proposed to learn procedure-aware representations, including learning spatiotemporal features [7, 18, 21, 77], using action labels as supervision [43, 50, 86, 102], incorporating temporal order of steps [28, 102], and consulting external activity procedure knowledge databases [104] such as WikiHow [31]. However, these approaches often struggle to explicitly model procedural knowledge that relies on how scene states evolve over time and how they could have evolved if the precondition and/or result of an action step had differed. For instance, the procedure “*Make Oatmeal*” in Figure 1 (c) cannot be finished correctly if the state of the bowl remains “*Full of oatmeal and without water*” or “*Full of water without oatmeal*” because neither state meets the precondition of the later action step, “*Heat with microwave*.” Capturing these subtle transitions requires *state-change reasoning* to understand how an action transforms the environment and objects, and *counterfactual analysis* to foresee alternative outcomes for the same action.

In this paper, we propose a hierarchical procedure-aware video representation learning framework that leverages state-changes and counterfactuals generated by a Large Language Model (LLM) [17]. We use the actual state-change descriptions to temporally contrast features, and the counterfactuals as the negative exemplars in contrastive learning [10, 29, 54] to enhance the modeling of clip-level and video-level procedure transformations. Specifically, at the clip level, we first model *before* and *after states* to describe the change in the environment and/or objects resulting from an action. We also incorporate *state-change counterfactuals* that estimate a hypothesized state if the action

step were to fail for any reason. For example, if the action were “make coffee,” a possible result state counterfactual could be “cup shards scattered on the floor.” We then use temporal contrastive learning to encourage the model to learn a representation space in which visual features of later frames in short clips and *after states* are close together, and in which the early frames, *before state*, and *state counterfactual* are far away, and vice versa. At the video level, we also generate *missing-step* and *misordered* counterfactuals that simulate the hypothesized scenarios by disturbing action steps within a long video. Then, we treat these video-level counterfactuals as hard negatives in video-language alignment, enabling a more comprehensive understanding of the entire procedure in a long-form video.

To validate our approach, we incorporate state changes and their counterfactuals into the learning process and pre-train our model on Ego4D [23, 42], a large-scale daily activity dataset filtered to include clean and aligned short clips along with their corresponding clip-level narration text and video-level summary text. We evaluate the learned procedure-aware video representations in six key procedural and short-term video understanding tasks—error detection, temporal action segmentation, multi-instance retrieval, action recognition, action phase classification, and frame retrieval—and show that they significantly enhance performance compared to strong baselines. In addition, we conduct a comprehensive ablation study to highlight the impact of learning state changes and their counterfactuals, demonstrating their importance in enhancing procedural video understanding. To summarize, our contributions are:

1. a novel approach to video representation learning that explicitly incorporates state changes and their counterfactuals to enhance procedure awareness;
2. a hierarchical learning framework that integrates state-change and counterfactual descriptions into frame-, clip-, and video-level feature alignment; and
3. experiments showing state-of-the-art performance on several procedure-aware video downstream tasks and comprehensive analysis and discussion.

2. Related Work

2.1. Video Procedure-Aware Representations

Procedure understanding, which involves recognizing and localizing individual action steps and reasoning about their causal relationships, is a fundamental challenge in computer vision, and is important for applications such as step recognition [5, 13, 76], temporal action segmentation [11, 20, 32, 64], and error detection [37]. Early work proposed learning procedure-aware video representations via exploiting the temporal consistency within videos [5, 55, 82, 86, 98] and temporal order [28] as pre-text tasks. Some work incorporates task graphs as guidance to learn procedure-aware rep-

representations, either by constructing graphs based on knowledge bases [104] or through learning [3, 6, 48, 52, 63]. Recently, several papers have attempted to align video features in datasets such as HowTo100M [49] or Ego4D [23] with text descriptions extracted through automatic speech recognition [48, 50], refined subtitles [43] with the external database WikiHow [31, 48], or manually labeled annotations [2, 42, 102]. These text descriptions offer rich context of action steps and temporal relations, opening the door for procedure-awareness learning. For example, Brprompt [39] generates text prompts from ground truth action labels to learn ordinal temporal features. PVRL [102] formulates future action steps as next-token prediction [14], enhancing temporal order between actions. HierVL [2] proposes hierarchical video-text alignment for both short clips and long videos.

However, these papers fall short of explicitly learning state changes—the causal relationships between action and state—which is essential to procedure understanding [53, 72]. Moreover, they can overfit to the seen correctly executed actions, which can hinder both understanding activity procedures and recognizing deviations such as erroneous action steps. In this work, we use state-change descriptions and counterfactuals to force our model to learn about action transformations and hypothetical failure outcomes, under the hypothesis that this type of learning will facilitate better procedure awareness.

2.2. State-Change Understanding

The idea of visual state changes has been explored extensively in the computer vision community, sometimes using different terms such as transformations between states [41, 83] or state-modifying actions [53, 70, 71, 73]. For example, papers have studied object state change localization [1, 45, 60, 70, 91] and generation [73], and action recognition with state changes [19, 60, 83]. Procedure planning, which aims to generate effective plans for task completion based on start and goal visual states, also incorporates the concept of understanding state changes [9, 40, 47, 51, 53, 78]. We were also inspired by SCHEMA [53] to use LLMs to generate before- and after-state descriptions of actions.

However, three key factors sets our work apart from this existing work. First, unlike existing work that mainly focuses on a specific task such as procedure planning [53], we aim to design a more general video representation learning framework for various downstream tasks. Second, prior work does not incorporate hypothetical outcomes, i.e., counterfactuals, which we believe are essential for procedure causal reasoning. Last but not least, prior work often only models state changes involving objects that are directly interacted with. In contrast, our generated descriptions can cover state changes of objects, humans, and the surrounding environment. We provide extensive examples in our supplementary materials.

mentary materials.

2.3. Video Representations with Counterfactuals

Counterfactual example reasoning has been shown to help models better understand cause-and-effect and predict outcomes [61, 92, 97], yielding strong generalization and robustness on tasks such as video question-answering [38, 95, 100] and compositional reasoning [35, 75]. Extensive research has studied creating counterfactual examples with image data augmentation [103], rule-based text transformations [67], random word substitutions [59], language models [16, 81, 88, 99], and synthetic image-text pairs generated by a simulator [8] or generative model [35, 36, 66]. However, existing work mainly focuses on creating counterfactuals for images or short videos and does not consider the procedural nature of activities. In this work, we propose state-change counterfactuals at both the clip and video levels, highlighting both action transformations and procedure evolution.

3. Proposed Method

We first present the text generation process for state changes and state-change counterfactuals. Then, we provide an overview of the clip-video hierarchical representation learning framework. Finally, we present our pretraining optimization that incorporates the generated texts to learn procedure-aware representations.

3.1. State Change & Counterfactual Generation

We use the Llama 3 [17] LLM to generate clip-level state descriptions (*before state*, *after state* and *state-change counterfactuals*) and to generate video-level summary counterfactuals (*missing step* and *misordered step*).

Clip-Level State Changes and their Counterfactuals.

We feed the narration annotated for each short clip in Ego4D [23, 42] into Llama and ask it to generate a state-change description based on that narration. Specifically,

Generate [Before] describing the scene before the action is performed and [After] describing the scene changed by the action.

For example, given the narration “*C picks a bag of clothes from the floor.*” where *C* is the camera-wearer, Llama might generate [Before]: “*The floor is cluttered with clothes*” and [After]: “*The bag of clothes is now in C’s hand, with the surrounding area slightly rearranged.*” Then, we generate multiple *state-change counterfactuals (SC-CF)* using another prompt with the same action context,

Now, based on the state changes you generated, create 3 *state-change*

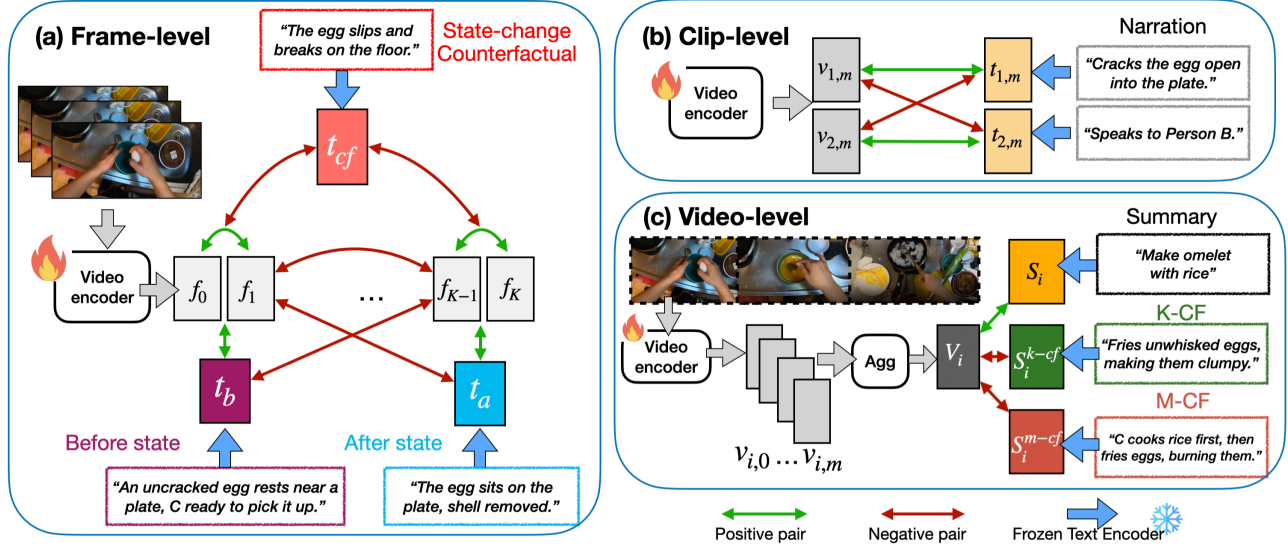


Figure 2. Illustration of our learning framework. (a) We train **frame** features by temporally contrasting neighboring and distant frames by incorporating **Before**, **After**, and **State-change Counterfactuals (SC-CF)**. (b) We align **clip** features with narration features. (c) Clip features are aggregated with the aggregator (Agg) as **video** features. Then the video features are contrasted positively with summaries and negatively with **Missing-step Counterfactuals (K-CF)** and **Misordered Counterfactuals (M-CF)**. Note that all text features are extracted with the frozen text encoder.

counterfactuals [SC-CF] depicting scenes with incomplete or incorrectly completed actions.

The generated results [SC-CF] might be “Clothes remain scattered on the floor” or “A small pile of clothes sits amidst remaining clutter.”

Video-Level Counterfactual Generation. We feed all clip narrations in a long video and their video-level summaries annotated in Ego4D [23, 42] to Llama. We ask the LLM to generate summary counterfactuals for *missing key steps* [K-CF] in which some crucial action steps (clip narrations) are missing,

Generate 10 distinct
counterfactuals [K-CF] by
taking out some key actions.

and *misordered* [M-CF] counterfactuals in which the order of key action steps is permuted,

Generate 10 distinct
counterfactuals [M-CF] by
perturbing the order of actions.

In this way, we generate diverse counterfactuals that can facilitate models to learn from hypothetical and unseen scenarios. As an example, consider the video summary “Make omelet with rice” in Figure 2. The procedure consists of a sequence of actions, [“Crack eggs,” “Whisk eggs,” “Fry eggs,” “Fry rice”, ...]. The video-level state-change counterfactuals could be missing action step “Whisk

eggs”, which might lead to state “Clumpy eggs”, while if one misordered the action step “Fry rice” by performing it too early, it could lead to state “Burnt rice.” We provide more examples of the generated descriptions in the supplementary materials.

3.2. Preliminary: Hierarchical Video Feature Learning

We adopt HierVL [2], a framework that learns hierarchical video-language representations at two temporal scales, clip-level and video-level. The clip-level narrations and video-level summaries are annotated by Ego4D/EgoClip [23, 42]. Note that since HierVL trains both a visual and text encoder, it requires vision-to-text and text-to-vision alignment. Below we describe the former, and the latter is defined symmetrically. Formally, let a video be divided into M short clips $\{v_1, \dots, v_M\}$ with corresponding narration texts $\{t_1, \dots, t_M\}$, and let S be the video-level summary for the entire video. HierVL’s vision-to-text training objective consists of two parts: clip-level alignment and video-level alignment losses.

Clip-Level Alignment (Child Loss) is a contrastive loss that encourages matched (v_i, t_i) pairs to have high similarity relative to mismatched pairs. Specifically, we use a softmax-based contrastive objective,

$$\mathcal{L}_{v2t} = \frac{-1}{|B|} \sum_{i \in B} \log \frac{\sum_{p \in P(i)} \exp(v_i^T t_p / \tau)}{\sum_{j \in B} \exp(v_i^T t_j / \tau)}, \quad (1)$$

where B is the mini-batch, v_i and t_j are visual and textual embeddings, respectively, and $P(i)$ denotes the positive samples of the i^{th} video clip. HierVL [2] leverages a variant of the action-aware loss, EgoNCE [42], in order to mine hard positives and negatives; please see [2, 42] for details. The overall loss at the child level is thus $\mathcal{L}_{\text{child}} = \mathcal{L}_{v2t} + \mathcal{L}_{t2v}$, where the latter is defined symmetrically.

Video-Level Alignment (Parent Loss) captures long-range context. Clip features $v_1, \dots, v_m$ and text features $t_1, \dots, t_m$ are aggregated into single video-level representations V_i and T_i , respectively,

$$\begin{aligned} V_i &= \text{Agg}(v_1, \dots, v_m), \\ T_i &= \text{Agg}(t_1, \dots, t_m), \end{aligned}$$

where $\text{Agg}(\cdot)$ is an aggregator function implemented with a self-attention transformer. This parent loss seeks to align the video-level features to the text summary provided by Ego4D,

$$\mathcal{L}_{\text{parent}}^{SV} = \frac{-1}{|B|} \sum_{i \in B} \log \frac{\sum_{p \in P(i)} \exp(V_i^T S_p / \tau)}{\sum_{j \in B} \exp(V_i^T S_j / \tau)}, \quad (2)$$

where the superscript SV emphasizes that this is the vision-to-text alignment loss, and $\mathcal{L}_{\text{parent}}^{ST}$ is defined analogously for text-to-text alignment. The overall loss at the parent level is thus $\mathcal{L}_{\text{parent}} = \mathcal{L}_{\text{parent}}^{SV} + \mathcal{L}_{\text{parent}}^{ST}$.

3.3. Pretraining Objective: State Change & Counterfactual

We use a hierarchical contrastive pre-training framework with three timescales: frame-, clip-, and video-level.

Frame-Level Alignment. At the frame level, the model is supervised by our proposed *before-state* loss $\mathcal{L}_{\text{before}}$ and *after-state* loss $\mathcal{L}_{\text{after}}$, each defined as [29],

$$\mathcal{L} = \frac{1}{|B|} \sum_{i \in B} \frac{-1}{|P(i)|} \sum_{z_p \in P(i)} \log \frac{\exp(f_i^T z_p / \tau)}{\sum_{z_n \in N(i)} \exp(f_i^T z_n / \tau)}, \quad (3)$$

where B is the batch size, f_i is the visual embedding of the i^{th} frame, z_j is either a visual or text embedding, τ is a temperature hyperparameter, and $P(i)$ and $N(i)$ denote the positive and negative samples of the i^{th} frame, respectively. Given a set of $K = 4$ frames sub-sampled from a video clip, $\mathcal{L}_{\text{before}}$ aims to align earlier-in-time frames along with the *before-state* text embeddings, while pushing them apart from later-in-time frames, the *after-state*, and counterfactual text embeddings. That is, $P(0) = \{f_1, t_b\}$ and $N(0) = \{f_3, t_a, t_{cf}\}$, where t_b , t_a , and t_{cf} denote the text embeddings corresponding to the before state, after state, and the counterfactuals, respectively. On the other hand, $\mathcal{L}_{\text{after}}$ aims to align later-in-time frames to the after

state while separating them from earlier-in-time frames, the before-state, and counterfactual text embeddings. In other words, $P(3) = \{f_2, t_a\}$ and $N(3) = \{f_0, t_b, t_{cf}\}$. Additionally, other samples from the same mini-batch are included as negatives in the denominator of Eq. (3) in both losses; we omit them in the notation above for simplicity.

Clip-Level Alignment. At the clip-level, we use the $\mathcal{L}_{v2t}$ loss described in Section 3.2, which seeks to align the *video-clip* embeddings to their corresponding text narrations from EgoClip [42]. Note that since we do not train a text encoder, the symmetric $\mathcal{L}_{t2v}$ loss is neglected here. For more details on this loss, see Section 3.2 and [2, 42].

The resulting loss for the first two scales is thus,

$$\mathcal{L}_{\text{child}} = \mathcal{L}_{v2t} + \lambda(\mathcal{L}_{\text{before}} + \mathcal{L}_{\text{after}}), \quad (4)$$

where λ is a hyperparameter controlling the strength of the state-change aware supervision.

Video-Level Alignment. The goal of this loss is to align video-level visual embeddings to summary text embeddings and to enhance procedural awareness using video-level counterfactuals. We first obtain video-level visual embeddings from clip-level embeddings using a self-attention block, as described in Section 3.2. Then, using contrastive learning, each visual embedding is aligned to its corresponding summary text embedding and contrasted against text embeddings of the generated counterfactual **K-CF** and **M-CF**. The formulation is defined as,

$$\mathcal{L}_{\text{parent}} = - \sum_{i \in B} \log \frac{\sum_{p \in P(i)} \exp(V_i^T S_p)}{\sum_{n \in N(i)} \exp(V_i^T S_n) + \exp(V_i^T S_{n,w}^{cf})}, \quad (5)$$

where V_i is the aggregated video-level visual embedding, S_j is a summary text embedding and $S_{j,w}^{cf}$ are **Misordered** and **Missing-step** counterfactual text embeddings where $w \in \{1, \dots, W\}$ is the total number of counterfactuals used, and $P(i)$ and $N(i)$ denote the positive and negative samples of the i^{th} video, respectively. We also use a temperature hyperparameter τ as in Eq. (3), but we omit it in the equation above for simplicity of notation. Note that consistent with [2], in this case, the summation over positive samples is located inside the logarithm since the sampling strategy of [42] is used. However, prior work [29] finds the summation outside the logarithm to be more effective and thus this is the de-facto choice of the frame-level loss in Eq. (3).

4. Experiments

4.1. Implementation Details

Pretraining. We pretrain our model on EgoClip [42], a dataset derived from Ego4D [23], with clean clip-level narrations and the video-level summaries. We follow [2, 42]

Table 1. Temporal action segmentation results on the GTEA and EgoPER datasets. Bold denotes best and underline is second best. **Text encoder** denotes the VLM model with a trainable text encoder. Following [44] we also present the average across all metrics (Avg).

Method	Pretraining Data	GTEA						EgoPER			
		F1@10	F1@25	F1@50	Edit	Acc	Avg	F1@50	Edit	Acc	Avg
Br-prompt [39]	GTEA	94.1	92.0	83.0	91.6	81.2	88.4	-	-	-	-
I3D [7]	Kinetics	<u>90.1</u>	<u>88.8</u>	79.2	84.6	<u>79.7</u>	84.5	48.8	71.9	73.9	64.9
CLIP [56]	WIT [74]+ Text encoder	88.5	<u>86.2</u>	77.6	87.1	<u>75.6</u>	83.0	44.2	71.2	70.8	62.1
MIL-NCE [50]	HowTo100M	67.9	61.3	44.6	67.9	58.3	60.0	47.3	69.1	73.6	63.3
PVRL [102]	HowTo100M	85.2	82.6	72.2	81.1	71.2	78.5	45.6	73.2	73.4	64.1
HierVL [2]	Ego4D+ Text encoder	90.4	88.5	<u>81.2</u>	86.7	78.5	<u>85.1</u>	<u>52.6</u>	<u>73.0</u>	<u>77.3</u>	<u>67.6</u>
Ours	Ego4D	89.8	89.1	81.6	<u>86.8</u>	80.0	85.5	54.4	74.1	79.0	69.2

to use the same training split that has 3.8M clip narrations and 120K long-term video summaries for pretraining. We pretrain our model on 8 NVIDIA L40S GPUs, with batch size 18 on each GPU for 7 epochs. We follow HierVL [2] and train models on clip- and video-level alignment alternatively. Specifically, we perform video-level alignment as in Eq. (5) after every 5 mini-batches of clip-level alignment as in Eq. (4). The frame-level loss is computed during the clip-level training iteration. We set the learning rate and weight decay to $1e-5$ and $1e-4$, respectively. Note that each short clip is sub-sampled to 4 frames during training [2, 42].

Text Generation. To generate all state-change descriptions and state-change counterfactuals, we use Llama 3.1 8B [17] for efficiency. We incorporate off-the-shelf text-encoder FLAVA [69] to pre-extract features of the annotated narrations and summaries and our generated text descriptions.

Temporal Action Segmentation. We evaluate representations on temporal action segmentation, in which the goal is to output frame-wise action labels for untrimmed videos. This requires modeling procedures and recognizing the start and end of state transformations. We evaluate various representations by adopting the popular temporal action segmentation model ASFormer [94] that takes video features as input. We follow prior work [39] for the same training strategy and evaluation protocol and metrics including F1 scores, Edit distance, and frame-wise Accuracy, on the cooking procedure datasets, Georgia Tech Egocentric Activities (GTEA) [20] and EgoPER [37] Dataset. GTEA consists of 28 egocentric instructional videos capturing daily kitchen activities at 15 frames per second. EgoPER contains 368 videos spanning 28 hours of cooking scenarios.

Error Detection. We evaluate representations on unsupervised error detection on the EgoPER dataset [37] where models are trained with error-free videos and output labels for each frame on whether it is an error. The error categories include *step omission*, *step addition*, *step modification*, *step slip*, and *step correction*. We follow [37], and use the EgoPED model [37] and equip it with differ-

ent video representations. Specifically, the training process of EgoPED consists of two stages: action segmentation and prototype learning. In stage 1, the temporal action segmentation model, ASFormer, is trained to predict frame-wise action labels on the error-free set by taking pre-extracted video features as input. In stage 2, the model learns prototype features for each class of action step, e.g., “*Take bowl from microwave*,” with contrastive step prototype learning [37, 54]. For each class of activity, multiple prototypes are extracted for error detection at inference. Note that each activity is trained independently, such as *Make Coffee*. During the inference, EgoPED calculates the similarity score of observed frame features and any learned prototypes. A threshold is further used to determine whether the observed test frame is erroneous.

Action Retrieval & Recognition. To demonstrate the applicability of the learned representations in core video understanding tasks, we evaluate zero-shot multi-instance retrieval on Epic-Kitchens (EK) [12] and zero-shot action recognition on Charades-Ego (CE) [68]. Following prior work [2, 42], both tasks are implemented via text-to-vision alignment by selecting the textual label with the highest similarity score to the given visual sample. Note that [42] reports overfitting when applying the model trained on Ego4D to Charades-Ego, and thus uses a task-specific checkpoint. We follow [2] and report the results of our pre-training checkpoint (denoted as PT ckpt in their work) rather than using task-specific checkpoints.

Action Phase Classification & Frame Retrieval. To assess *fine-grained* and *short-term procedure awareness*, we test representations on action phase classification and zero-shot frame retrieval on the Align-Ego-Exo (AE2) dataset [90]. The AE2 dataset contains egocentric and exocentric videos of actions *Break Eggs*, *Pour Milk*, *Pour Liquid*, and *Tennis Forehand*. Each action is divided into two to four phases [90]. We follow [90] and train an SVM to predict the per-frame action phase labels for the classification task and use nearest neighbors for the zero-shot retrieval

Table 2. Error detection results on the EgoPER dataset. Bold indicates best and underline is second best. HTM denotes the HowTo100M dataset [49]. **Text** denotes the VLM model with a trainable text encoder.

Method	Pretraining Data	Quesadilla		Oatmeal		Pinwheel		Coffee		Tea		All	
		EDA	AUC	EDA	AUC	EDA	AUC	EDA	AUC	EDA	AUC	EDA	AUC
Random	-	19.9	50.0	11.8	50.0	15.7	50.0	8.20	50.0	17.0	50.0	14.5	50.0
HF ² -VAD [46]	-	34.5	62.6	25.4	62.3	29.1	52.7	10.0	59.6	36.6	62.1	27.1	59.9
S3R [85]	Kinetics+I3D	52.6	51.8	47.8	61.6	50.5	52.4	16.3	51.0	47.8	57.9	43.0	54.9
I3D [7]	Kinetics	62.7	65.6	51.4	65.1	59.6	55.0	55.3	58.3	56.0	66.0	57.0	62.0
CLIP [56]	WIT [74]+ Text	77.6	67.2	69.6	67.5	<u>66.9</u>	<u>59.5</u>	68.5	<u>69.0</u>	75.6	57.7	<u>71.6</u>	<u>64.2</u>
MIL-NCE [50]	HTM	77.3	59.8	69.8	61.5	65.7	53.0	68.0	67.9	69.8	61.5	70.1	60.7
PVRL [102]	HTM	75.7	<u>70.0</u>	<u>71.2</u>	53.5	65.5	65.5	67.5	65.4	<u>76.4</u>	65.2	71.3	63.9
HierVL [2]	Ego4D+ Text	<u>77.9</u>	70.2	70.8	<u>66.4</u>	65.2	58.1	67.4	69.8	75.1	<u>66.4</u>	71.3	66.2
Ours	Ego4D	78.9	63.7	71.6	46.1	68.3	53.8	<u>68.3</u>	68.0	76.6	70.9	72.7	60.5

task. Note that we merge the validation and test sets for more robust results.

Baselines. We evaluate the popular I3D and CLIP feature [7, 56] commonly used in long-form video tasks and procedure-aware representations with publicly available pretrained model weights, including MIL-NCE [50], Bridge-prompt [39], and PVRL [102]. In addition, we evaluate the VLM HierVL [2] with only its video encoder.

4.2. Temporal Action Segmentation Results

Table 1 presents results on temporal action segmentation. The first line of the table shows Br-prompt [39] which is pre-trained on the target dataset (GTEA) and thus serves as an upper bound on performance. In our results, we found that MIL-NCE performs significantly worse than others, especially on GTEA. We assume this is because MIL-NCE is trained with low-quality automatic-speech-recognition (ASR) text and imprecise alignment between video and ASR sentences, unlike PVRL [2] which leverages pseudo-labels generated by CLIP [56]. We further compare our model with the VLM HierVL. Even though HierVL is trained with both video and text encoders, which require considerably more computational resources, our model outperforms HierVL in most metrics on both datasets. Furthermore, the performance gap against all non-VLM models is larger. This highlights the effectiveness of our proposed state-change descriptions and their counterfactuals on long-term procedure understanding.

4.3. Error Detection Results

Table 2 presents results of the error detection benchmark on EgoPER [37]. We first include results reported in EgoPER as a reference in the top group (rows 1-3) instead of a direct comparison. In the remaining groups (rows 4-9), we report the performance of representation learning approaches, tested using the same downstream architecture [94]. We ob-

Table 3. Multi-Instance Retrieval (MIR) on the Epic-Kitchens (EK) dataset [12] and Zero-shot Action Recognition on the Charades-Ego (CE) dataset [68].

Method	MIR (EK)		Action Rec. (CE)
	mAP	nDCG	mAP
CLIP [56]	8.4	<u>15.3</u>	<u>20.5</u>
MIL-NCE [50]	5.8	10.3	6.9
PVRL [102]	6.1	11.4	7.9
Ours	15.7	22.6	24.8

serve that all procedure-aware representations outperform general visual representations, such as I3D and CLIP, highlighting the importance of procedure awareness in the context of error detection. Furthermore, our proposed method surpasses the state-of-the-art in EDA [37], even outperforming the VLM HierVL.

Moreover, we achieve competitive performance on the AUC metric. Note that although we follow prior work [37] to include frame-wise metric AUC, this metric may not be effective in measuring the performance and the procedure awareness of representations; as discussed in EgoPER [37], a heuristic random method can achieve competitive results in AUC but fails to localize erroneous segments according to EDA. This is also true when comparing I3D and earlier methods, such as HF²-VAD [46] and S3R [85]. We hope our extensive experimental results can inspire future work to study more effective evaluation metrics for error detection. In summary, our representation learning method achieves state-of-the-art performance in EDA and competitive performance in AUC, highlighting the strong effectiveness of *counterfactual reasoning* in procedure awareness.

4.4. Action Retrieval & Recognition Results

Table 3 presents benchmark results on action retrieval and recognition. We found existing procedure-aware represen-

Table 4. Action Phase Classification and zero-shot Frame Retrieval results on the Align-Ego-Exo dataset [90].

Method	Classification (F1)		Retrieval (mAP)	
	ego+exo	ego	ego+exo	ego
CLIP [56]	59.5	61.6	64.4	68.0
MIL-NCE [50]	53.0	54.2	59.5	63.0
PVRL [102]	59.4	62.7	61.6	66.6
Ours	61.3	64.8	64.9	70.3

Table 5. Ablation study involving: **Before** & **After** states, **State-change Counterfactuals (SC-CF)**, **Missing-step (K-CF)** and **Misordered (M-CF)** on GTEA for temporal action segmentation and EgoPER for error detection. Note that we calculate EDA by averaging all categories of activity. Metrics: Acc. and EDA; mean $\pm$ sd of 5 and 10 runs, respectively.

ID	Before & After	SC-CF	K-CF	M-CF	GTEA (AS)	EgoPER (ED)
1 (Weak HierVL)					77.3 $\pm$.6	71.2 $\pm$.5
2	✓				78.4 $\pm$.4	72.1 $\pm$.6
3	✓	✓			79.1 $\pm$.2	71.5 $\pm$.8
4	✓	✓	✓		78.7 $\pm$.7	72.2 $\pm$.4
5	✓	✓		✓	78.8 $\pm$.8	72.2 $\pm$.5
Ours	✓	✓	✓	✓	79.6$\pm$.4	72.3$\pm$.5

tations perform significantly worse on *short-term* action recognition tasks, while our representations trained with both short- and long-term state-changes and counterfactuals work well on popular video recognition tasks and datasets.

4.5. Action Phase Classification & Retrieval Results

Table 4 shows that our model outperforms others on average across actions on both the *ego+exo* and *ego* settings, demonstrating its effectiveness in *short-term procedures* and strong view-point generalization. Interestingly, CLIP [56] demonstrates strong performance. We conjecture this is due to the focus on short-term action in its image-to-text alignment. Full results with individual actions are available in the supplementary material.

4.6. Ablation Study

In Table 5, we study the effectiveness of state changes (**Before** and **After**), state-change counterfactuals (**SC-CF**), and video-level state-change counterfactuals (**Missing-step CF** and **Misordered-step CF**) by ablating each of them and evaluating on temporal action segmentation with GTEA and error detection with EgoPER. Note that the first row (ID 1) can be seen as *Weak HierVL* as it is trained with only the video encoder. In addition, since *Weak HierVL* only uses narrations from other videos in a batch as negatives, it can be seen as a *random description* approach. All ablations (including *Ours*) are pre-trained with a batch size of 12 on each GPU.

Temporal Action Segmentation. One interesting insight

on GTEA is that using only clip-level state changes and their counterfactuals (ID 3) achieves competitive performance, outperforming most other ablated models. Also, using only one type of video-level counterfactual (ID 4 and 5) may mislead the model to overfit to a certain type of counterfactual, resulting in lower performance. By considering all the proposed states and counterfactuals together, we show that they complement each other, thus achieving the best performance.

Error Detection. The results on EgoPER show that our proposed state changes and their counterfactuals can improve performance by combining them all together, verifying the effectiveness of recognizing erroneous labels. In addition, we observe interesting results showing that with only clip-level counterfactuals (ID 3), the model performs slightly worse than its baseline (ID 2). We conjecture this is because the model overfits to short-term procedures and overlooks long-term activity procedures and their counterfactual reasoning.

5. Conclusions

In this work, we present a novel procedure-aware video representation learning framework that first incorporates state-change descriptions and state-change counterfactuals in clip-level alignment, enhancing causal reasoning of action transformations. Then, it uses video-level counterfactuals that perturb the local actions and create hypothesized scenarios to facilitate the understanding of activity procedures. Our learned representations demonstrate strong effectiveness in terms of procedure awareness and achieve state-of-the-art results on several benchmarks. For future work, a promising direction may be to generate multi-modal counterfactual examples for procedure-aware activities, such as synthetic long-form videos showcasing mistakes and the corresponding corrections.

6. Acknowledgment

Authors CK, FR, JH, and DC were supported in part by the National Science Foundation under award DRL-2112635 to the AI Institute for Engaged Learning and by the IU Luddy Artificial Intelligence Center. YC was supported in part by the National Science and Technology Council under grants 113-2628-E-A49 -022 - and 114-2628-E-A49 -007 -, by the Higher Education Sprout Project of National Yang Ming Chiao Tung University, and by the Ministry of Education through the Yushan Fellow Program Administrative Support Grant. Computation resources were supported in part by the Lilly Endowment, Inc. through its support to the IU Pervasive Technology Institute.

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